

Session 3: Calibration of the NextGen Rainfall-Runoff Model

Step-by-step commands for calibrating the NextGen rainfall-runoff model using the `ngiab-cal` tool for USGS gage 05454300 (Coralville). Two workflows are documented: a native Linux/conda environment and a Docker container. Each workflow completes calibration; the Docker workflow additionally includes validation.

Workflow A: Native Linux (conda environment)

These steps assume you are inside the activated conda environment.

Step 1: Extract data for calibration

Run `ngiab-cal` against the Coralville dataset to generate the calibration configuration:

```
uvx ngiab-cal /workspace/data/Coralville/ -g 05454300 -i 5 -w 365 --cr 0.5
```

Flag reference for this command:

- `/workspace/data/Coralville/` : path to the subset dataset produced in Session 2.
- `-g 05454300` : USGS gage ID used as the calibration reference.
- `-i 5` : number of calibration iterations.
- `-w 365` : warm-up period in days.
- `--cr 0.5` : calibration ratio (fraction of the simulation period allocated to calibration; the remainder is used for validation).

Step 2: Launch the calibration container

Start the `ngiab-cal:custom` container, mounting the Coralville dataset into the expected `/ngen/ngen/data` location and running as the current user so output files are owned correctly:

```
docker run -it -v "/workspace/data/Coralville/:/ngen/ngen/data" \
--user $(id -u):$(id -g) ngiab-cal:custom
```

Step 3: Start calibration

Inside the container, launch the calibration workflow:

```
./run.sh
```

Workflow B: Docker container

This workflow runs calibration and validation from inside the `ngiab-cal:custom` container. Because the container references its own default paths in the generated configuration files, several `sed` edits are required to redirect them to the Coralville dataset before calibration and validation are launched.

Step 1: Generate the calibration configuration

Run `ngiab-cal` on the Coralville dataset to produce the calibration files:

```
uvx ngiab-cal /workspace/data/Coralville/ -g 05454300 -i 5 -w 365 --cr 0.5
```

Step 2: Update paths in the calibration configuration files

The generated configuration files reference the container's default data location (/ngen/ngen/data). Redirect them to the actual Coralville path, and update the calibration launch script to point at the correct calibration directory:

```
sed -i 's|/ngen/ngen/data|/workspace/data/Coralville|g' \
/workspace/data/Coralville/calibration/ngen_cal_conf.yaml
sed -i 's|/ngen/ngen/data|/workspace/data/Coralville|g' \
/workspace/data/Coralville/calibration/realization.json
sed -i 's|/ngen/ngen/data|/workspace/data/Coralville|g' \
/workspace/data/Coralville/calibration/troute.yaml
sed -i 's|/workspace/data/calibration|/workspace/data/Coralville/calibration|' \
/calibration/calibrate.sh
```

Step 3: Run calibration

Launch the calibration workflow:

```
/calibration/calibrate.sh
```

Step 4: Update paths to use the Coralville validation configuration

Point validate.sh at the validation configuration produced under the Coralville directory:

```
sed -i \
's|/workspace/data/calibration/Output/Validation_Run/ngen_cal_conf.yaml|/workspace/data/Coralville/calibration/Output/Validation_Run/ngen_cal_conf.yaml|g' \
/calibration/validate.sh
```

Redirect both the container default data path and the default calibration directory inside the validation YAML:

```
sed -i \
-e 's|/ngen/ngen/data|/workspace/data/Coralville|g' \
-e 's|/workspace/data/calibration|/workspace/data/Coralville/calibration|g' \
/workspace/data/Coralville/calibration/Output/Validation_Run/ngen_cal_conf.yaml
```

Apply the same two path replacements to the validation realization JSON:

```
sed -i \
-e 's|/ngen/ngen/data|/workspace/data/Coralville|g' \
-e 's|/workspace/data/calibration|/workspace/data/Coralville/calibration|g' \
/workspace/data/Coralville/calibration/Output/Validation_Run/realization.json
```

The validation script contains a hard-coded reference to the original realization file. Replace that line so it points at the Coralville realization:

```
sed -i \
's|^[[:space:]]*original_realization = Path(.*)|          original_realization = Path("/workspace/data/Coralville/calibration/realization.json")|' \
/calibration/ngen-cal/python/runCalibValid/validation.py
```

Step 5: Run validation

With all paths redirected, launch the validation workflow:

```
/calibration/validate.sh
```

Running NextGen with the Calibrated Forcing

Once calibration is complete, the calibrated parameters can be applied to a NextGen simulation over a new period. The steps below prepare a fresh dataset (Coralville2, 2020-01-01 to 2021-01-01) and launch a NextGen run using the calibrated forcing.

Workflow A: Docker container

Step 1: Prepare a new dataset for the NextGen run

Generate subset, forcings, and realization for a new simulation window:

```
uvx ngiab-prep -i gage-05454300 -sfr \  
  --start_date 2020-01-01 --end_date 2021-01-01 \  
  -o Coralville2 --source aorc
```

Step 2: Launch the NextGen simulation

From inside the Coralville2 directory, run the NextGen entry-point script, passing the current directory as the dataset path:

```
cd Coralville2  
bash /ngen/HelloNGEN.sh $(pwd)
```

Workflow B: Native Linux (conda environment)

Step 1: Enter the NGIAB-CloudInfra directory

```
cd /home/msmg8/NGIAB-CloudInfra/
```

Step 2: Launch the guided NextGen run

Start the interactive guide script, which walks through the dataset selection and NextGen execution:

```
./guide.sh
```